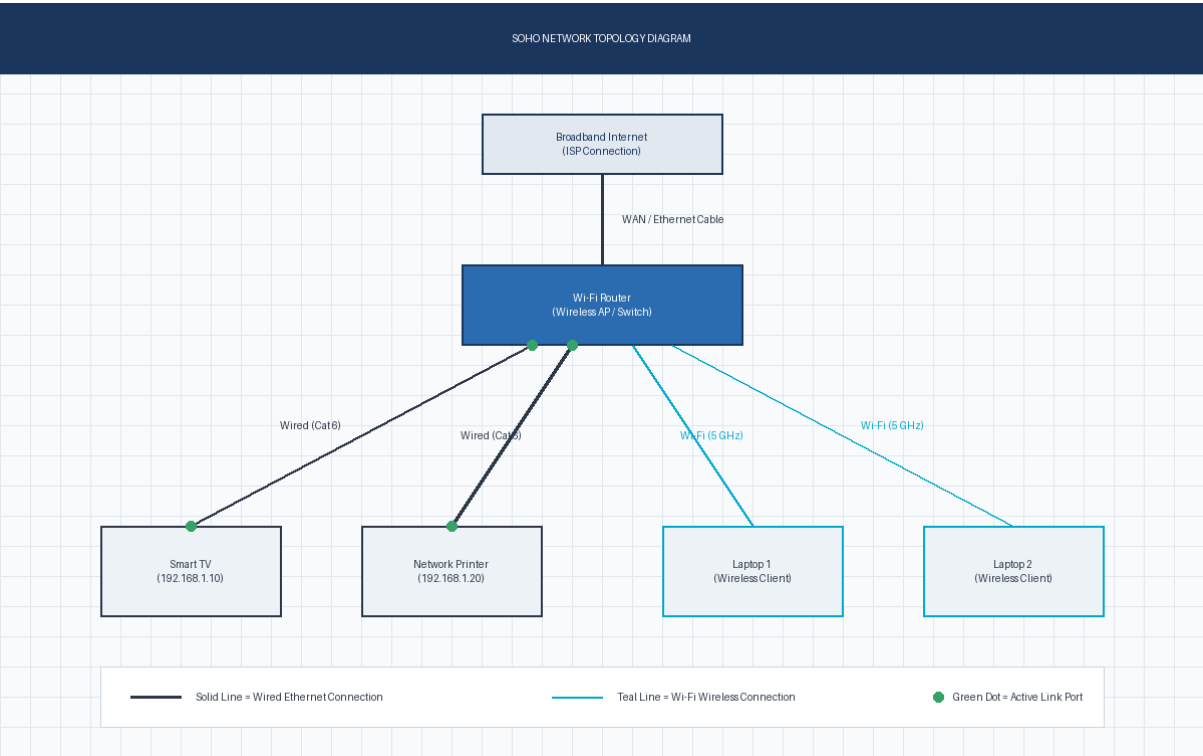


TASK-1

SOHO NETWORK DIAGRAM



TASK-2

To build the physical SOHO (Small Office/Home Office) network from the topology diagram, here is the complete hardware bill of materials (BOM) with recommended, readily available models (commonly found on Amazon India, Flipkart, and local retailers).

 Hardware Components & Equipment List

1. Central Network & Routing Equipment

Component	Function	Model Suggestion	Connectivity Port
Fiber ONT / Modem	Converts ISP fiber signal into Ethernet	Syrotech / Digisol GPON ONT (Usually provided by ISP like Airtel Xstream or JioFiber)	1x Optical Port, 1x RJ45 Gigabit WAN

Component	Function	Model Suggestion	Connectivity Port
Wi-Fi 6 Router	Main gateway, DHCP server, & dual-band AP	TP-Link Archer AX12 / AX10 (AX1500) or D-Link DIR-X1560	1x Gigabit WAN Port, 4x Gigabit LAN Ports

2. End-User & Peripheral Devices

Device	Primary Access Type	Suggested Model Examples	Connection Detail
Smart TV	Wired (LAN)	LG 43-inch 4K UHD Smart TV or Samsung Crystal 4K	Plugs into Router LAN Port 1 via Ethernet cable.
Network Printer	Wired (LAN)	HP Smart Tank 585 or Canon imageCLASS MF271dn	Plugs into Router LAN Port 2 via Ethernet cable.
Laptops (x2)	Wireless (Wi-Fi)	HP Pavilion / Dell Inspiron / Lenovo IdeaPad	Connects to the 5 GHz Wi-Fi SSID generated by the router.

3. Cabling & Network Accessories

- Cat6 RJ45 Ethernet Patch Cords (x3):**
 - Brand:** Amazon Basics, Fedus, or CableCreation Cat6 Ethernet Cables
 - Lengths Required:**
 - 1x Short Patch Cord (1 meter):** Connects Fiber Modem to Wi-Fi Router (WAN Port).
 - 2x Long Patch Cords (3m to 10m depending on room layout):** Connects Router LAN ports to the Smart TV and Printer.
- Unmanaged Gigabit Switch (Optional / Expansion):**
 - Model:** TP-Link TL-SG1005D (5-Port Gigabit Desktop Switch)
 - Use Case:** Useful if your router has fewer than 4 LAN ports or if you plan to add extra wired devices (like a desktop PC or NAS) later.

- **Power & Cable Management:**
 - **Surge Protector:** Belkin 4-Socket Surge Protector to power the modem, router, and printer safely.
 - **Cable Clips / Sleeves:** To neatly route cables along walls or under desks.



Wiring Setup Instructions

1. **WAN Link:** Connect an Ethernet patch cord from the **LAN port of the ISP Fiber Modem** to the **WAN (Internet) port of the TP-Link Router**.
2. **TV Link:** Connect a Cat6 Ethernet cable from **LAN Port 1** of the router into the Ethernet (RJ45) port on the back of the Smart TV.
3. **Printer Link:** Connect a Cat6 Ethernet cable from **LAN Port 2** of the router into the Ethernet (RJ45) port of the network printer.
4. **Laptop Setup:** Connect Laptop 1 and Laptop 2 wirelessly by scanning the Router's **5 GHz Wi-Fi network** and entering your secure WPA2/WPA3 password.

TASK-3

Step-by-Step: Configure a Wi-Fi Router for a SOHO Network

Step 1: Connect the Router

1. Connect the broadband/ISP cable to the router's **WAN (Internet)** port.
2. Connect your laptop to the router using an **Ethernet cable** or the default Wi-Fi network.
3. Turn on the router and wait for the power and Internet LEDs to become stable.

Step 2: Log In to the Router

1. Open a web browser (Chrome, Edge, etc.).
2. Enter the router's default IP address in the address bar:
3. `http://192.168.10.1`

(Some routers use 192.168.0.1 or 192.168.1.1.)

4. Enter the default **username** and **password** provided on the router label or in the manual.

Step 3: Change the Wi-Fi Network Name (SSID)

1. Go to **Wireless Settings**.
 2. Find the **SSID (Wi-Fi Name)** field.
 3. Enter a unique name, for example:
 4. Jaydeep_HomeWiFi
 5. Click **Save** or **Apply**.
-

Step 4: Enable WPA2 Security

1. In **Wireless Security**, select:
 - **Security Mode: WPA2-Personal (AES)**
2. Enter a strong Wi-Fi password, for example:
3. Home@12345
4. Click **Save** or **Apply**.

Note: WPA2 encryption helps protect the wireless network from unauthorized access.

Step 5: Change the Default Admin Password

1. Go to **Administration** or **System Tools**.
2. Select **Change Password**.
3. Enter the current admin password.
4. Create a new strong password, for example:
5. Admin@2026!
6. Confirm the new password and click **Save**.

Important: Changing the default admin password prevents unauthorized users from accessing the router's settings.

Step 6: Restart the Router

1. Restart the router if prompted.
 2. Reconnect your devices using the new Wi-Fi name (SSID) and password.
-

Step 7: Verify the Configuration

- Check that the new **SSID** is visible on your devices.
- Connect using the new **WPA2 password**.
- Open a web browser to confirm Internet access.
- Log in to the router again using the **new admin password**.

Configuration Summary

Setting	Example Value
Router IP Address	192.168.10.1
SSID	Jaydeep_HomeWiFi
Security Mode	WPA2-Personal (AES)
Wi-Fi Password	Home@12345
Admin Username	admin
New Admin Password	Admin@2026!

TASK-4

SOHO Network IP Configuration

Router Configuration

- Router IP Address: 192.168.10.1
- Subnet Mask: 255.255.255.0
- DHCP Range: 192.168.10.100 – 192.168.10.200

Device IP Assignment

Device	Connection Type	IP Assignment	MAC Address (Example)	Assigned IP Address
Wi-Fi Router	Wired	Static	00:1A:2B:3C:4D:01	192.168.10.1
Laptop 1	Wireless	DHCP	08:16:3E:5A:10:21	192.168.10.100

Device	Connection Type	IP Assignment	MAC Address (Example)	Assigned IP Address
Laptop 2	Wireless	DHCP	08:16:3E:5A:10:22	192.168.10.101
Smart TV	Wireless	Static	3C:52:82:9F:11:33	192.168.10.20
Printer	Wired	Static	A4:5E:60:7B:8C:44	192.168.10.10

Additional Network Settings

Setting	Value
Network Address	192.168.10.0/24
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	8.8.8.8 (Primary), 8.8.4.4 (Secondary)

Why These Assignments?

- **Printer:** Uses a static IP (192.168.10.10) so all devices can always find it for printing.
- **Smart TV:** Uses a static IP (192.168.10.20) for consistent streaming and easier management.
- **Laptop 1 & Laptop 2:** Use DHCP, allowing the router to automatically assign available IP addresses whenever they connect to the network.

Summary

- **Static IP Devices:** Router, Printer, Smart TV
- **DHCP Devices:** Laptop 1, Laptop 2
- **All devices are on the same subnet:** 192.168.10.0/24
- **Default Gateway for all devices:** 192.168.10.1

TASK-5

Troubleshooting a Laptop That Cannot Connect to the Internet in a SOHO Network

If one of the laptops in my SOHO network cannot access the internet, I would troubleshoot it using the following steps.

Possible Cause	How to Diagnose	How to Fix
1. Wi-Fi is disconnected or wrong password	Check whether the laptop is connected to the correct Wi-Fi network (SSID). Verify the Wi-Fi signal strength and password.	Reconnect to the correct SSID and enter the correct WPA2 Wi-Fi password. Move closer to the router if the signal is weak.
2. Incorrect IP Address or DHCP Issue	Open Command Prompt and run ipconfig to check the IP address. If the IP starts with 169.254.x.x, the laptop did not receive an IP from the router.	Run ipconfig /release followed by ipconfig /renew, or restart the router. Ensure DHCP is enabled on the router.
3. Router or Internet Connection Problem	Check whether other devices (such as another laptop or a smartphone) can access the internet. Check the router's Internet/WAN LED.	Restart the router and modem. If all devices are affected, contact the Internet Service Provider (ISP).
4. Incorrect Network Settings	Verify the default gateway and DNS server using ipconfig /all.	Set the default gateway to 192.168.10.1 and use valid DNS servers such as 8.8.8.8 or obtain them automatically.
5. Firewall or Antivirus Blocking Access	Temporarily disable the firewall or antivirus and test the connection.	Adjust the firewall or antivirus settings to allow internet access, then re-enable security protection.

Conclusion

To troubleshoot a laptop that cannot connect to the internet, I would first check the Wi-Fi connection, then verify the IP configuration using ipconfig, and finally ensure the router and internet connection are working properly. By checking each possible cause one by one, I can quickly identify the problem and restore internet connectivity.